



Sidharth Vijayan Krishnan

Software Engineer

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- 26/09/1993
- Indian
- Full working permission for Germany

EDUCATION

M.Sc. Computer Science
Technische Universität Clausthal
01/2021 – 09/2025
Clausthal-Zellerfeld, Germany

B.Tech. Computer Science and Engineering
Mar Baselios Institute of Technology and Science (MBITS)
08/2011 – 05/2015 | Kerala, India

LANGUAGES

- English** – Fluent (C1)
- German** – Intermediate (A2)

TECHNICAL SKILLS

- Languages** – Java, Python, JavaScript, SQL
- Backend** – Spring Boot, REST APIs, Node.js
- Frontend** – React, HTML5, CSS3, AngularJS
- Databases** – PostgreSQL, MySQL, SQL Server, Oracle
- Cloud & DevOps** – Git, Docker, Linux, CI & CD, GitHub Actions, Jenkins, Maven, Kubernetes, Bash/Shell
- AI & Data** – Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, TensorFlow, Jupyter
- Testing** – JUnit, Selenium, TestNG, JMeter, TDD/BDD, API Testing
- Concepts & Practices** – OOP, Data Structures & Algorithms, Agile/Scrum, Jira/Confluence

INTERESTS

- Hiking and outdoor activities |
- Photography | Cycling

EXPERIENCE

TU Clausthal

- Master's Researcher (Topic: Agent-Based Simulation of Emergency Evacuation in University Lecture Halls)
09/2024 – 09/2025 | Clausthal-Zellerfeld, Germany
- Designed an agent-based simulation to analyze emergency evacuation dynamics and crowd flow.
 - Modeled up to 200 concurrent agents with varied behaviors using a NetLogo simulation.
 - Simulated psychological factors across 5 lecture hall designs to map human decision-making.
 - Automated 9,000+ simulation runs via NetLogo BehaviorSpace for extensive parameter sweeps.
 - Engineered Python pipelines (Pandas, Matplotlib) to process and visualize evacuation metrics.
 - Proved that optimized seating and aisle structures reduce egress time by 15-20%.

Rogersoft Technologies Pvt. Ltd.

- Test Automation Engineer Intern
12/2018 – 03/2019 | Kerala, India
- Deployed automated CI/CD testing frameworks to accelerate web and mobile software delivery.
 - Automated 150+ manual test cases (Selenium, TestNG), reducing regression testing time by 40%.
 - Achieved 85% automated test coverage for core iOS/Android flows using Appium.
 - Integrated test suites into Jenkins, GitLab CI, and GitHub Actions, cutting feedback loops to under 2 hours.
 - Resolved 3 critical performance bottlenecks by simulating 5,000+ concurrent users in JMeter.

Matrix Technologies Pvt. Ltd.

- Academic Coordinator
11/2015 – 11/2018 | Kerala, India
- Delivered training in Java, OOP, data structures, and algorithms, equipping students with practical software development skills.
 - Guided student projects through the full SDLC, including requirements analysis, system design, SQL database integration, testing, and deployment.
 - Created hands-on coding exercises, technical documentation, and assessment materials to support learning outcomes and project success.
 - Coordinated academic operations across five departments, managing schedules, examinations, curriculum updates, and performance reporting.

Verbicio Labs Pvt. Ltd.

- Full-Stack Developer Intern
07/2015 – 09/2015 | Kerala, India
- Built responsive UI components (AngularJS, Bootstrap), improving user engagement by 15%.
 - Optimized SQL database schemas to securely manage 10,000+ user records.
 - Delivered the project MVP 1 week ahead of schedule within a cross-functional Agile team.
 - Resolved 30+ full-stack bugs to ensure a stable production environment release.

PROJECTS

- Master Project | Agent-Oriented Programming: Designing BDI Agents in GOAL**
03/2024 – 08/2024
- Engineered autonomous cognitive agents to solve team exploration challenges in a dynamic hex grid.
 - Programmed autonomous BDI cognitive agents using GOAL for complex environmental reasoning.
 - Developed decentralized algorithms for task distribution and navigation without a central controller.
 - Designed communication protocols to share spatial knowledge and optimize collective mapping.
 - Formulated logic-based action rules for pathfinding and decision-making under time constraints.